Loss of Plasticity in Deep Continual Learning

DOHARE, ET AL
NATURE 632, 2024

Continually Learning Agents Seminar - Winter Semester 2026
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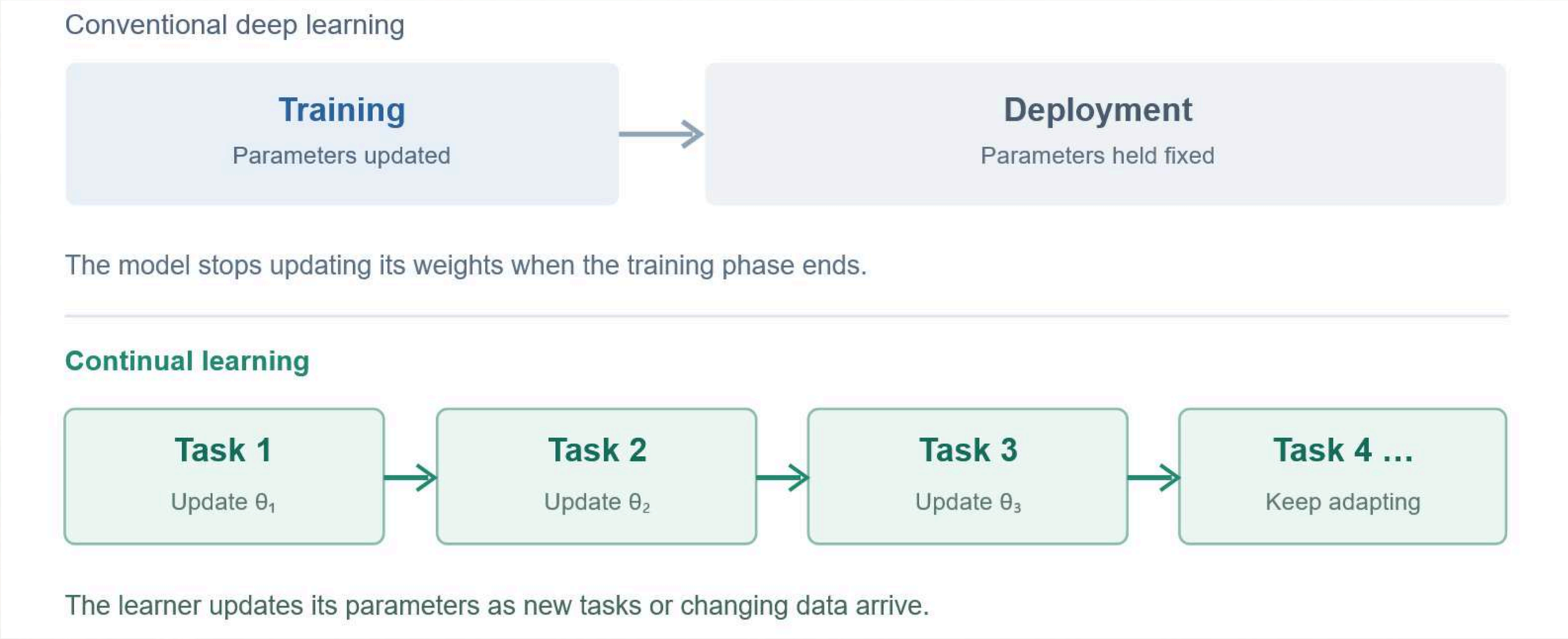
Assumptions & limitations

1.Introduction and motivation

After learning a 100 tasks, can a neural network still learn the 101st?



Continual learning requires adaptation beyond a fixed training phase



The paper's three objectives

01 — DEMONSTRATE
GRADUAL LOSS OF LEARNING ABILITY

02 — INVESTIGATE
INTERNAL CHANGES CORRELATED WITH THAT LOSS

03 — MITIGATE
INTRODUCE CONTINUAL BACKPROPAGATION TO HELP PRESERVE THE NETWORK'S CAPACITY TO LEARN.

2.Background & Related Work

Catastrophic forgetting vs Loss of plasticity

Catastrophic forgetting

Can the network still perform old tasks?

Stability of previously learned knowledge

Loss of plasticity

Can the network still learn new tasks?

Ability to acquire new knowledge

Plasticity is the ability to improve performance when learning from new data.

High Plasticity

Rapid improvement on a new task

Low plasticity

Limited improvement despite continued training

Two research traditions converged in the study of plasticity loss

Generating and selecting features

1958/59 Selfridge's Pandemonium Competing feature detectors

1969 Klopf & Gose's pattern-recognition network

2013 Mahmood & Sutton's Representation search through generate-and-test

Evidence of lost adaptability(losing ability to modify features to learn new info)

2000 Ellis & Lambon Ralph: Age-of-acquisition effects

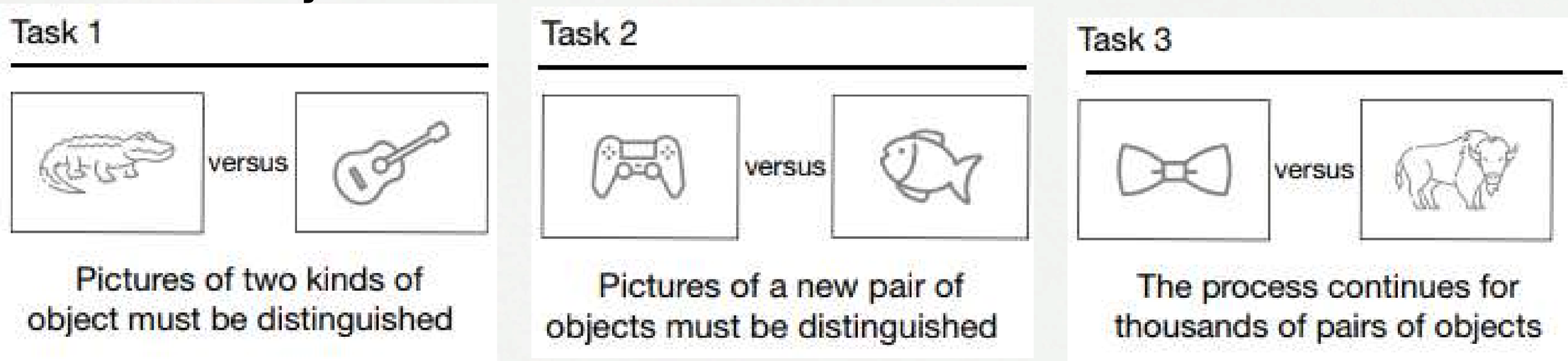
2018 Achille et al.: critical learning periods

2022–23 Lyle, Abbas, Sokar: capacity loss and dormancy

3. Demonstrating loss of plasticity

Experiment 1

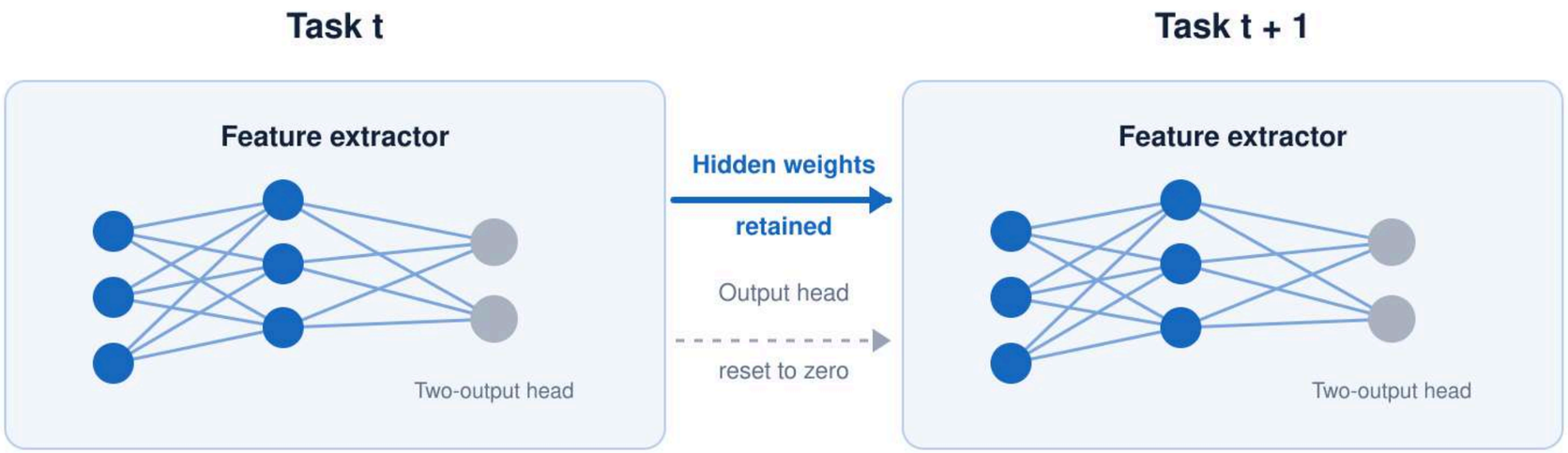
Continual ImageNet transforms image recognition into a sequence of binary classification tasks



The question is whether the same hidden representation remains trainable

SOURCE: DOHARE ET AL., NATURE 632 (2024), FIG. 1A

Hidden features carry forward while the two-output head resets



Train 250 epochs → Evaluate → Next class pair → Repeat

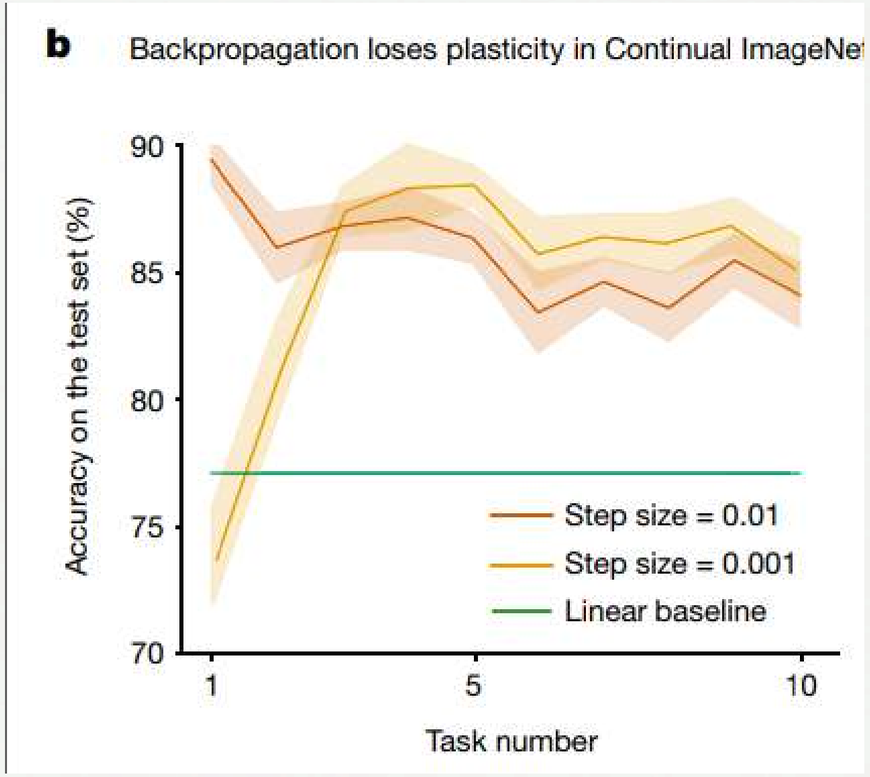
32 × 32	1,200	200	250
Image resolution	Training images per task	Test images per task	Epochs per task

A linear reference tests whether deep features still add value

Deep neural network	Linear baseline
Learns nonlinear features	No hidden feature extractor
Hidden weights persist across tasks	All learned weights reset
Accumulates learning history	Starts fresh on each task

Both models are evaluated on the current task; not on all past tasks

Backpropagation initially learns ImageNet pairs effectively

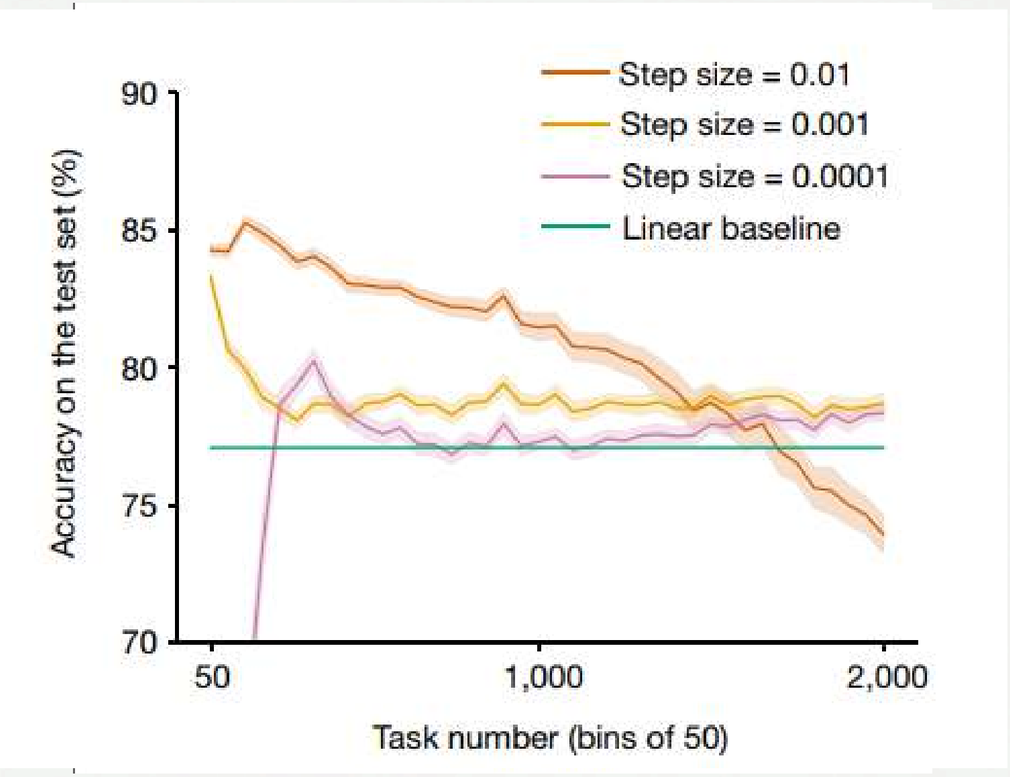


X-axis: first 10 tasks
Y-axis: test accuracy on each current task

Deep feature learning works at the start and accuracy reaches roughly 88% on early tasks

Source: Dohare et al., Nature 632 (2024), Fig. 1b, left

After 2,000 tasks, standard backpropagation loses its advantage



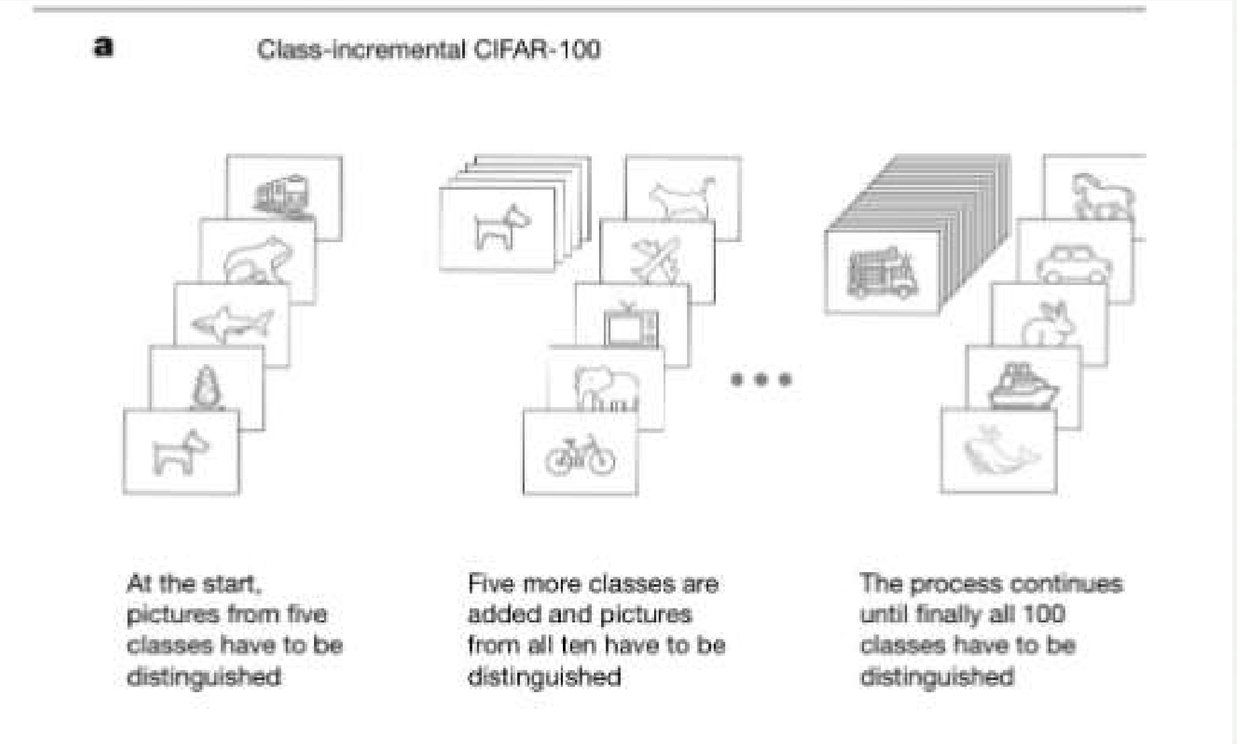
X-axis: task number (bins of 50)
Y-axis: current-task test accuracy
The decline persists across tested step sizes

The same model can become less capable of learning new tasks, even fall below linear baseline

Source: Dohare et al., Nature 632 (2024), Fig. 1b, right

Experiment 2

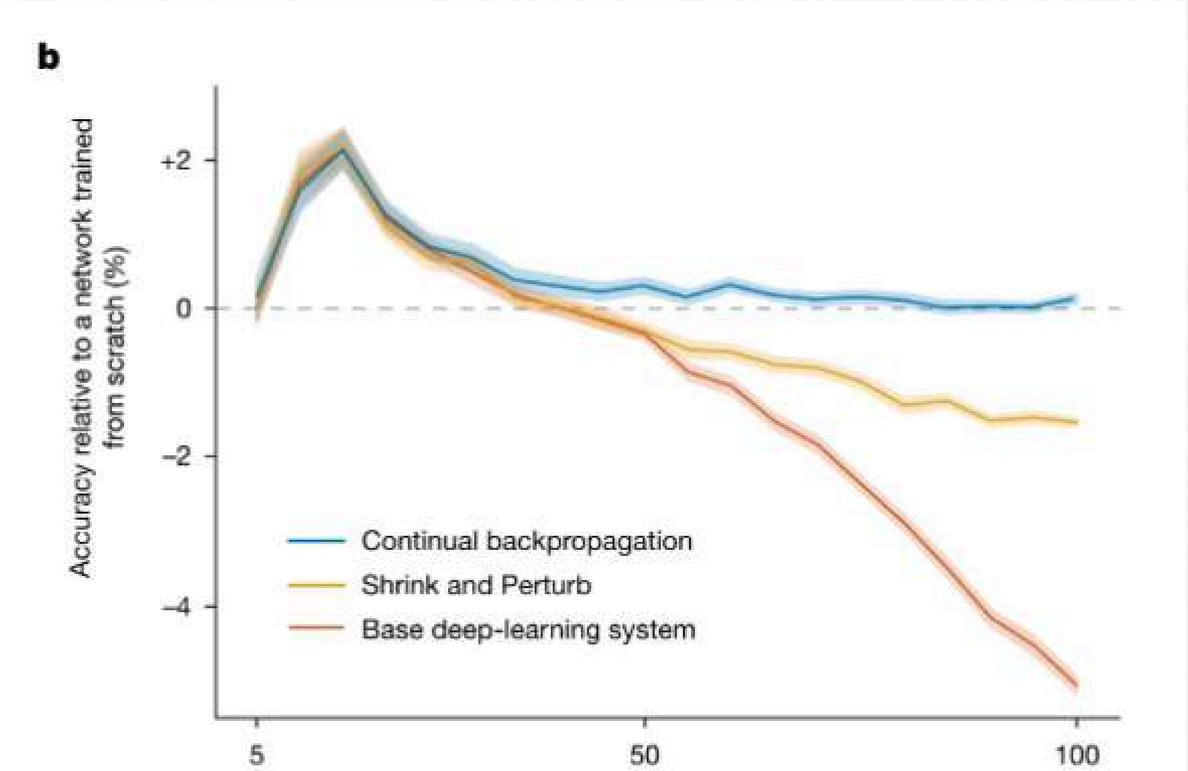
CIFAR-100 adds classes while retaining access to earlier data



Five new classes per increment · Earlier classes remain in training

Source: Dohare et al., Nature 632 (2024), Fig. 2a

Incremental training eventually falls behind fresh retraining



READ THE ZERO LINE

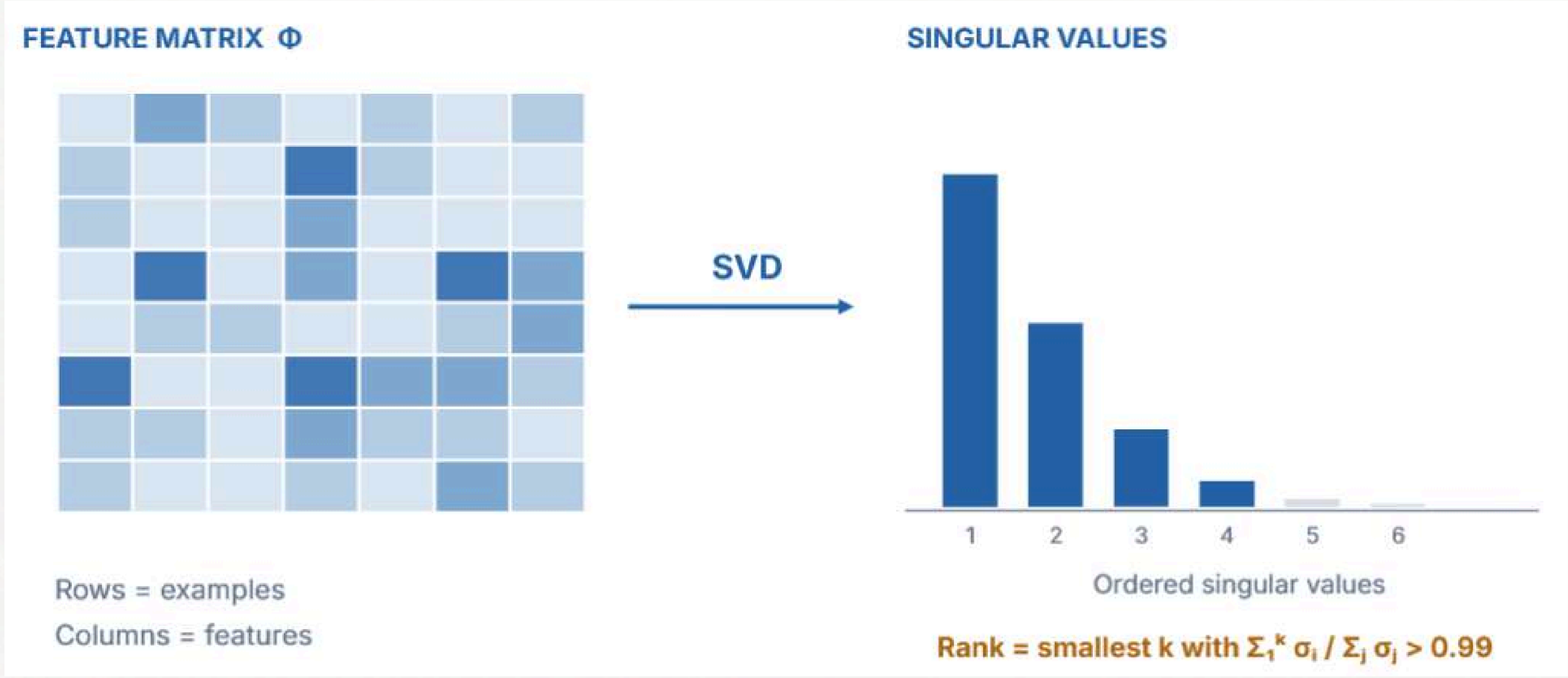
Above 0
Incremental training wins

Below 0
Fresh retraining wins

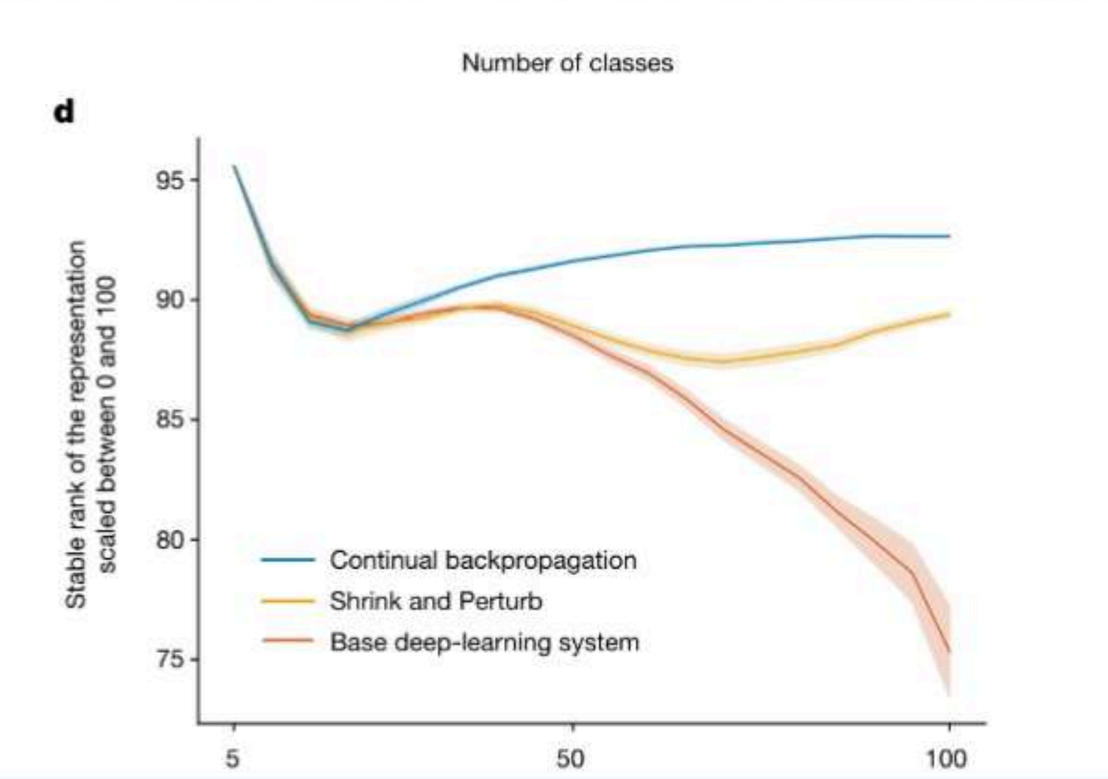
We will come to CBP

Source: Dohare et al., Nature 632 (2024), Fig. 2b

Representation rank measures the diversity of learned features



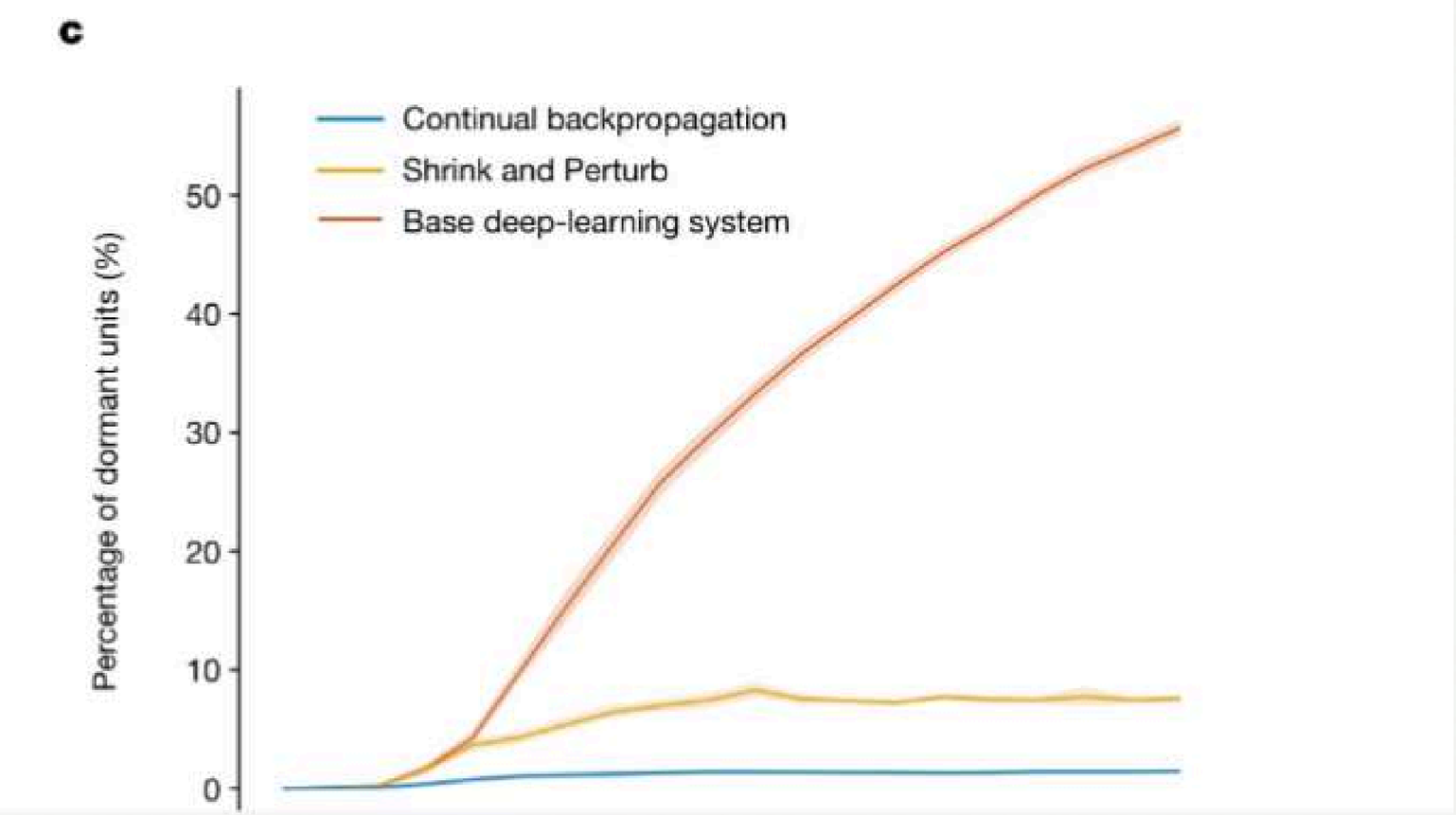
The base network progressively loses representation diversity



Lower stable rank means fewer effective representation directions

Source: Dohare et al., Nature 632 (2024), Fig. 2d

Dormant units accumulate in the base CIFAR-100 network



DORMANT ≠ DELETED

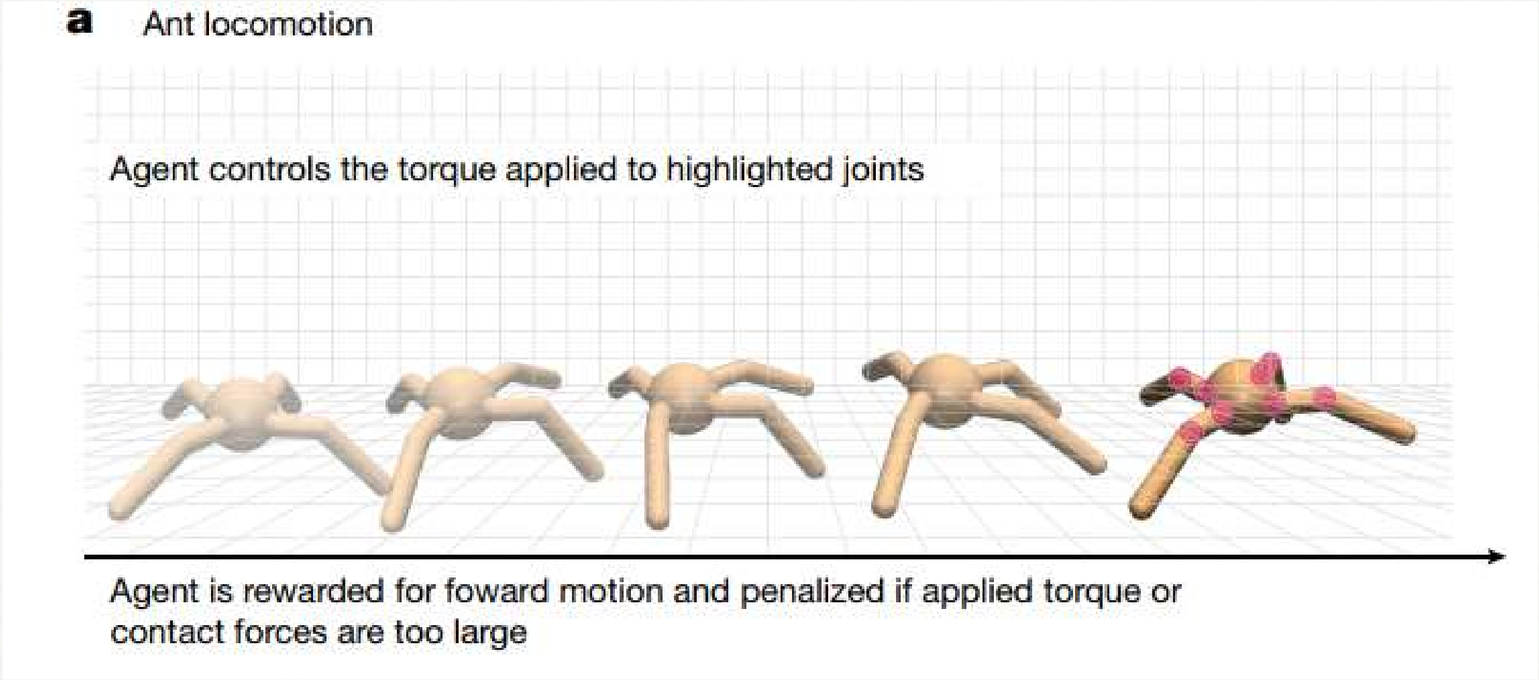
Very low observed activation on relevant data

The architecture stays the same, but fewer features contribut

Source: Dohare et al., Nature 632 (2024), Fig. 2c

Experiment 3

PPO learns an ant policy from actions, observations, and rewards



Observe
currrent state

→

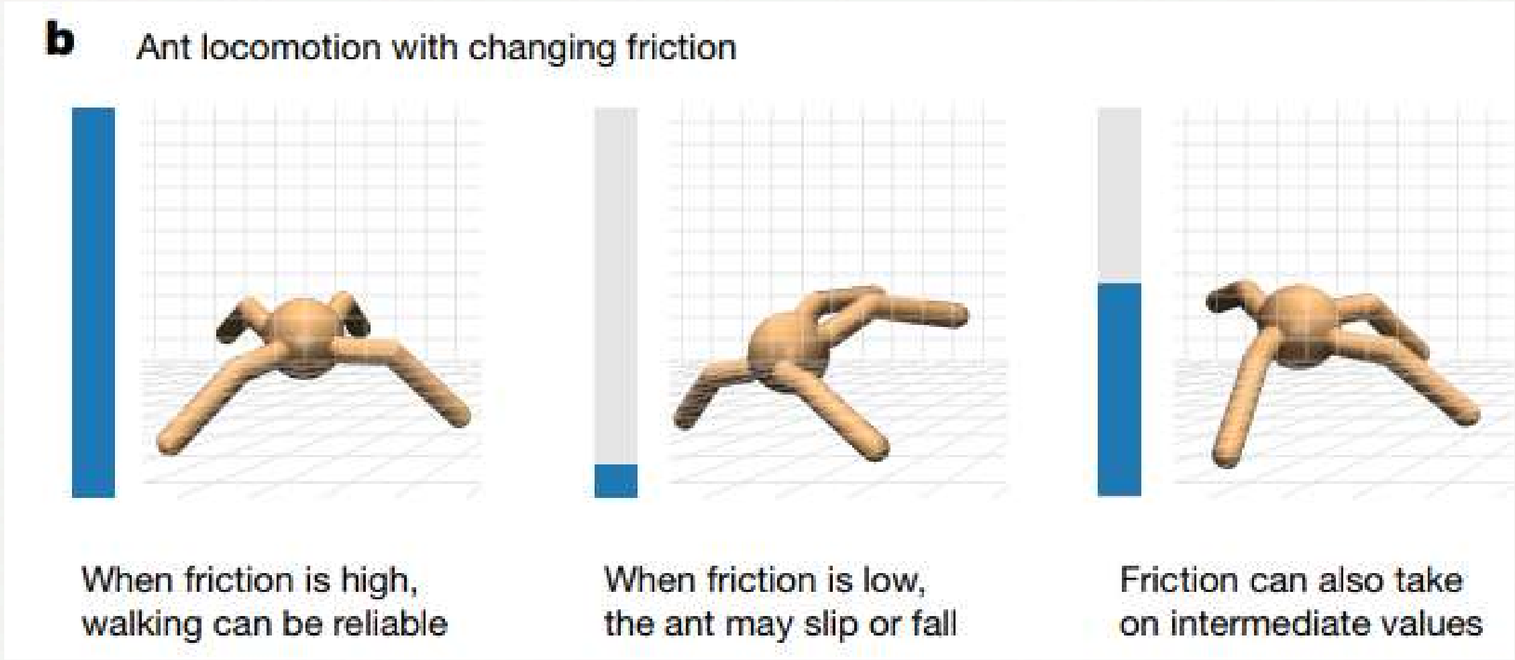
Act
Joint torques

→

Learn
Reward → PPO update

Source: Dohare et al., Nature 632 (2024), Fig. 3a

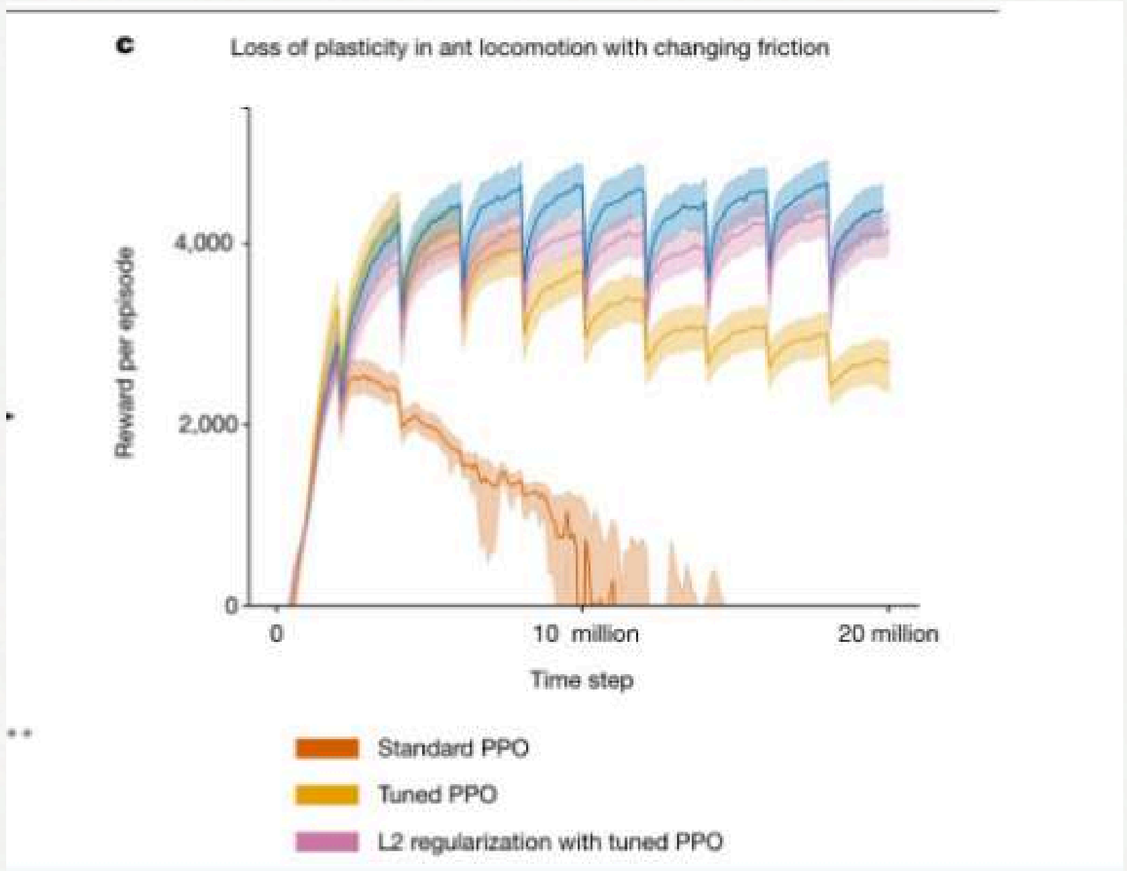
Changing friction forces the ant to repeatedly adapt



Ground friction changes every 2 million environment steps

Source: Dohare et al., Nature 632 (2024), Fig. 3b

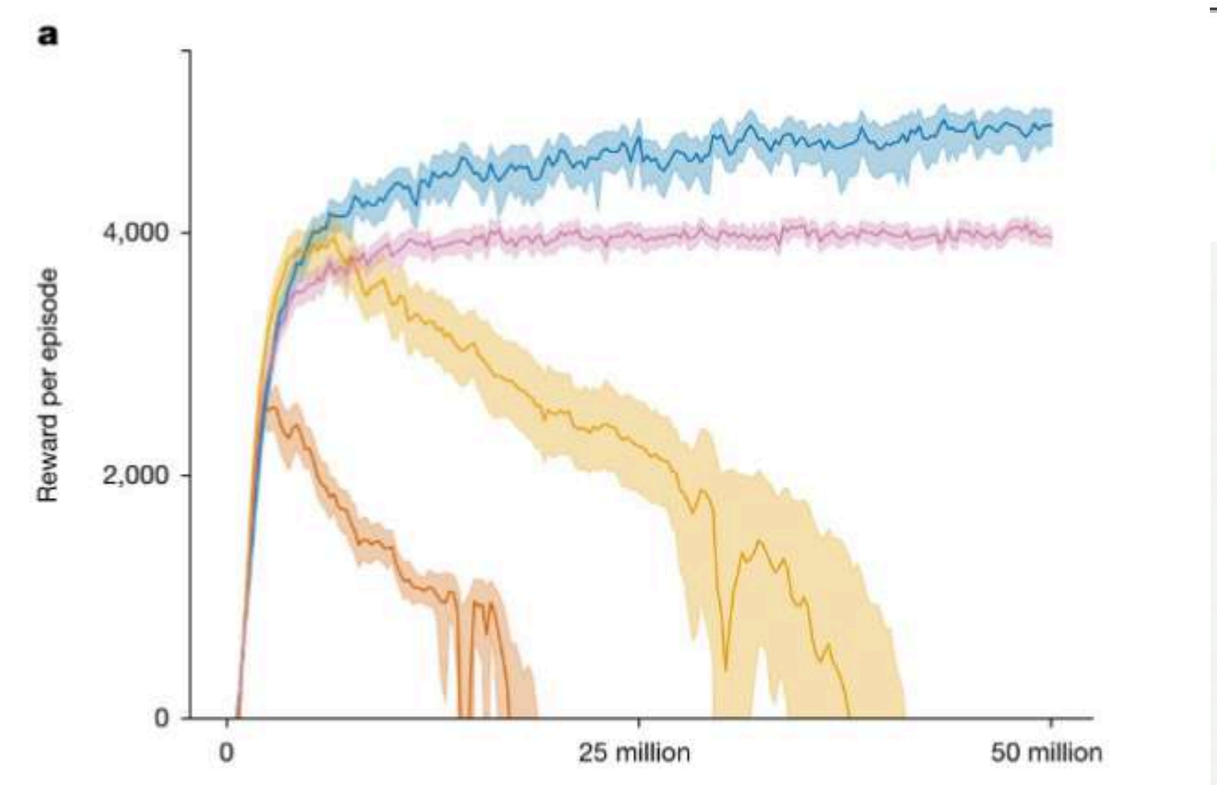
Standard PPO struggles to recover across repeated friction changes

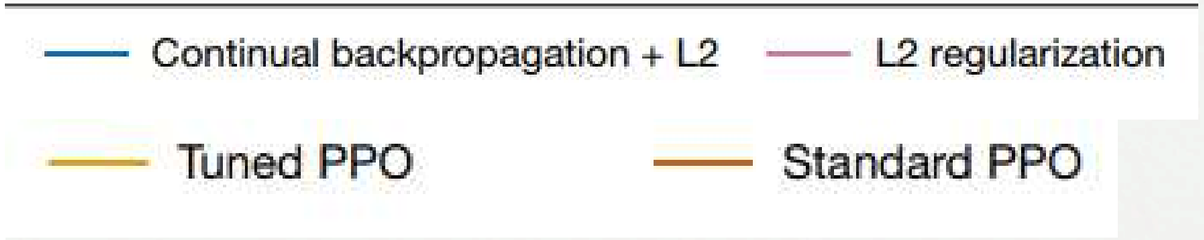


X · environment steps
Y · reward per episode
Dips · friction changes

Source: Dohare et al., Nature 632 (2024), Fig. 3c

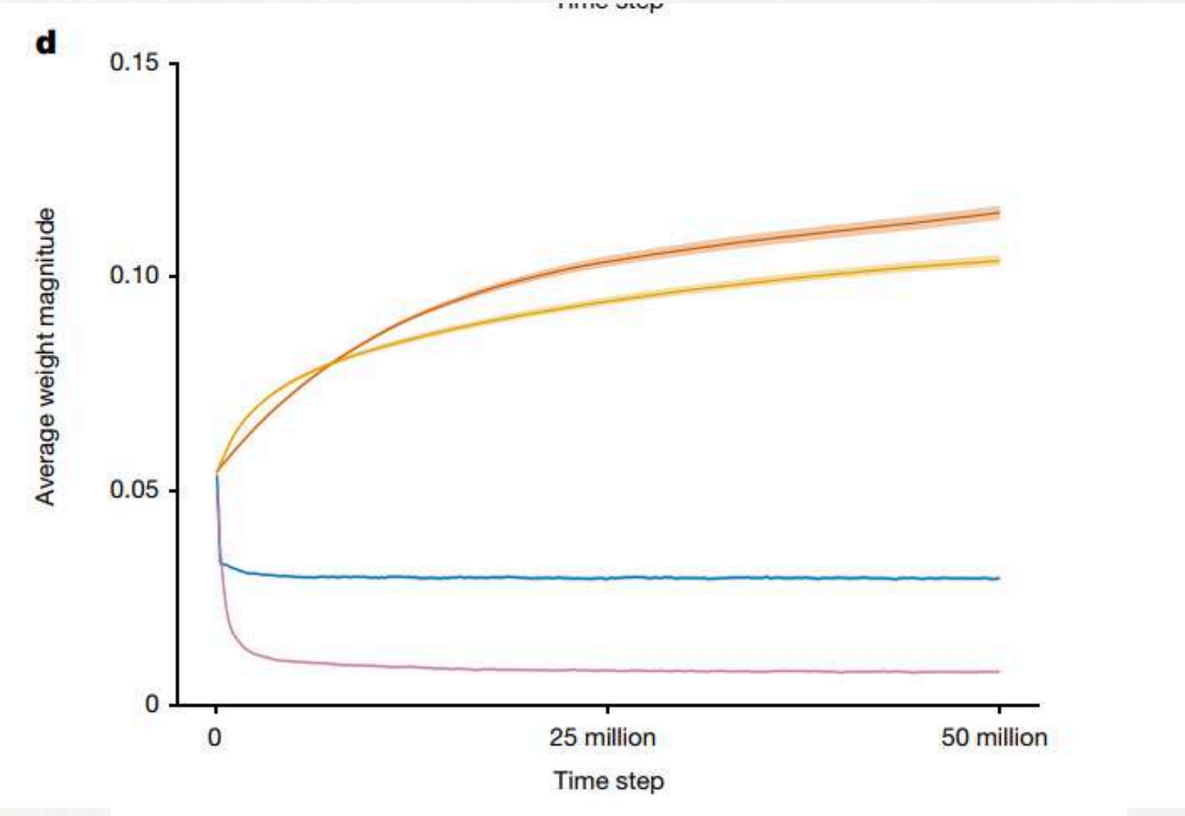
Standard PPO can collapse even with constant ground friction





Source: Dohare et al., Nature 632 (2024), Fig. 4a

Unregularized PPO develops increasingly large weights



Weight control becomes one candidate intervention

Source: Dohare et al., Nature 632 (2024), Fig. 4d

Three internal changes accompany plasticity loss

Dormant units increase

A growing fraction of units contributes little to current data

Representation rank decreases

Less effective feature diversity

Weight magnitudes increase

Parameters grow larger

4.Maintaining plasticity

The observed changes suggest two strategies for maintaining plasticity.

Weight growth → Control weight magnitudes
L2 Regularization

Reduced feature diversity → Maintain variability
Random purturbation/feature renewal

L2 regularization discourages weight growth during training

$$L_{\text{total}} = L_{\text{task}} + \lambda \|w\|^2$$

Task performance

Penalty on large weights

w — trainable weight λ — regularization strength

Shrink and Perturb combines weight control with continual random variation

Shrink
Move weights towards zero

Perturb
Add small random changes

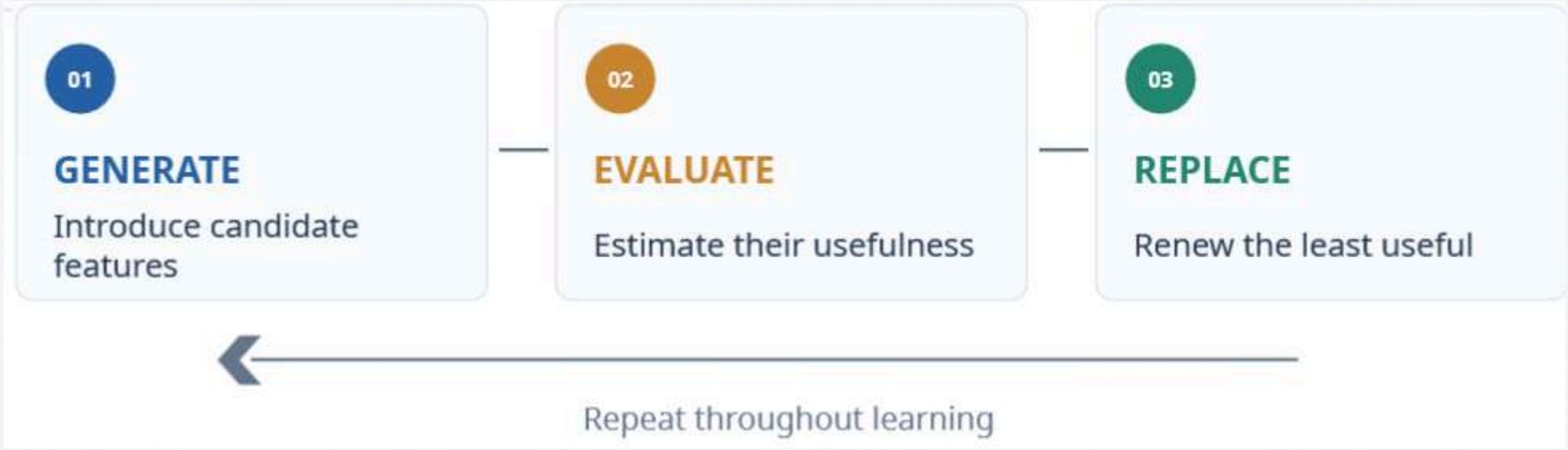
$$w' = \gamma w + \varepsilon$$

γ: shrinkage factor

ε: random perturbation

Source: Jordan et al., NeurIPS (2020)

Generate-and-test methods motivate selective replacement of low-utility features



Selfridge (1958/1959) → Mahmood & Sutton (2013) → Continual Backpropagation (2024)

CBP combines gradient learning with selective feature renewal

01 Standard backpropagation

Update weights to reduce the current task loss

Gradient-based learning

02 Selective renewal

Replace a mature, low-utility neuron

A small fraction of units per update, not the whole network

Repeat throughout training — learning and renewal happen together

Contribution utility estimates which neurons are least useful

Running contribution utility - Paper equation (1)

$$\mathbf{u}_l[i] = \eta \times \mathbf{u}_l[i] + (1-\eta) \times |\mathbf{h}_{l,i,t}| \times \sum_{k=1}^{n_{l+1}} |\mathbf{w}_{l,i,k,t}|, \qquad (1)$$

MEMORY

$\eta u_l[i]$ — previous utility

ACTIVITY

$|h_{l,i,t}|$ — unit activation

CONTRIBUTION

$\Sigma\ |w_{l,i,k,t}|$ — outgoing weights

$\eta = 0.99$

l: layer · i: neuron · t: update · k: next-layer neuron · n_{l+1}: next-layer width

Selection rule: among eligible units, replace the one with the lowest utility

Maturity and replacement rate prevent excessive neuron resets

MATURITY THRESHOLD m

New units are protected until age > m

REPLACEMENT RATE ρ

Expected replacements per step neligibleρ
Example: 512 × 10−5 = 0.00512 units per update.
Approximately one replacement per 195 updates

CBP reinitializes incoming weights and zeros outgoing weights of the selected neuron

Incoming weights	Outgoing weights
Reinitialize from the original random distribution	Set to zero, initially preserving the learned network output

Zero outgoing contribution → immediate prediction unchanged

The replaced unit’s utility and age also reset to zero.

A CBP step alternates learning and selective replacement

1. Run forward and backward passes; update weights

2. Update hidden-unit ages and utilities

3. Select the least useful mature unit when due

4. Reinitialize its connections, utility, and age

Core idea: gradients refine existing features; selective resets create new ones

CBP demo


Continual Backpropagation | Interactive Algorithm Walkthrough
cbp_algorithm_walkthrough
tiiny.site

5.Experimental evaluation

CBP is evaluated across three continual-learning settings

01 Supervised

Continual ImageNet

Current-task accuracy over 5,000 binary task

02 Incremental

CIFAR-100

RELATIVE ACCURACY, DORMANCY, AND STABLE RANK

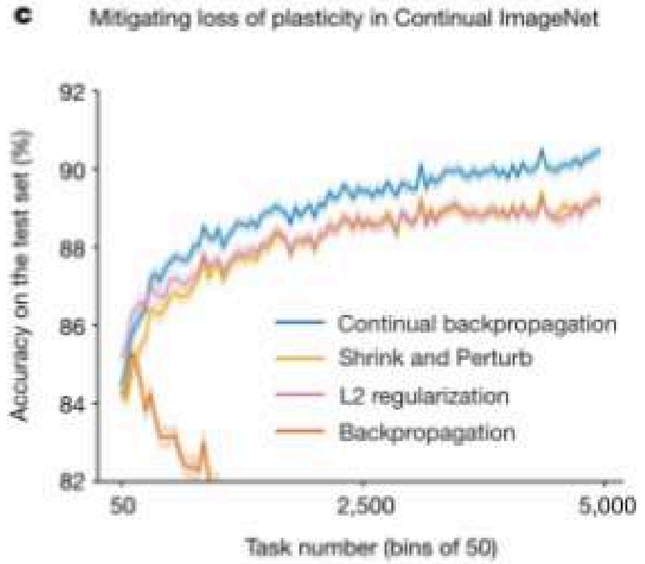
03 Reinforcement

ANT LOCOMOTION

REWARD UNDER CHANGING AND CONSTANT FRICTION

Experiment 1

CBP preserves learning performance on Continual ImageNet



Compared methods

- CBP
- Shrink and Perturb
- L2 regularization
- Backpropagation

X: task number, in bins of 50

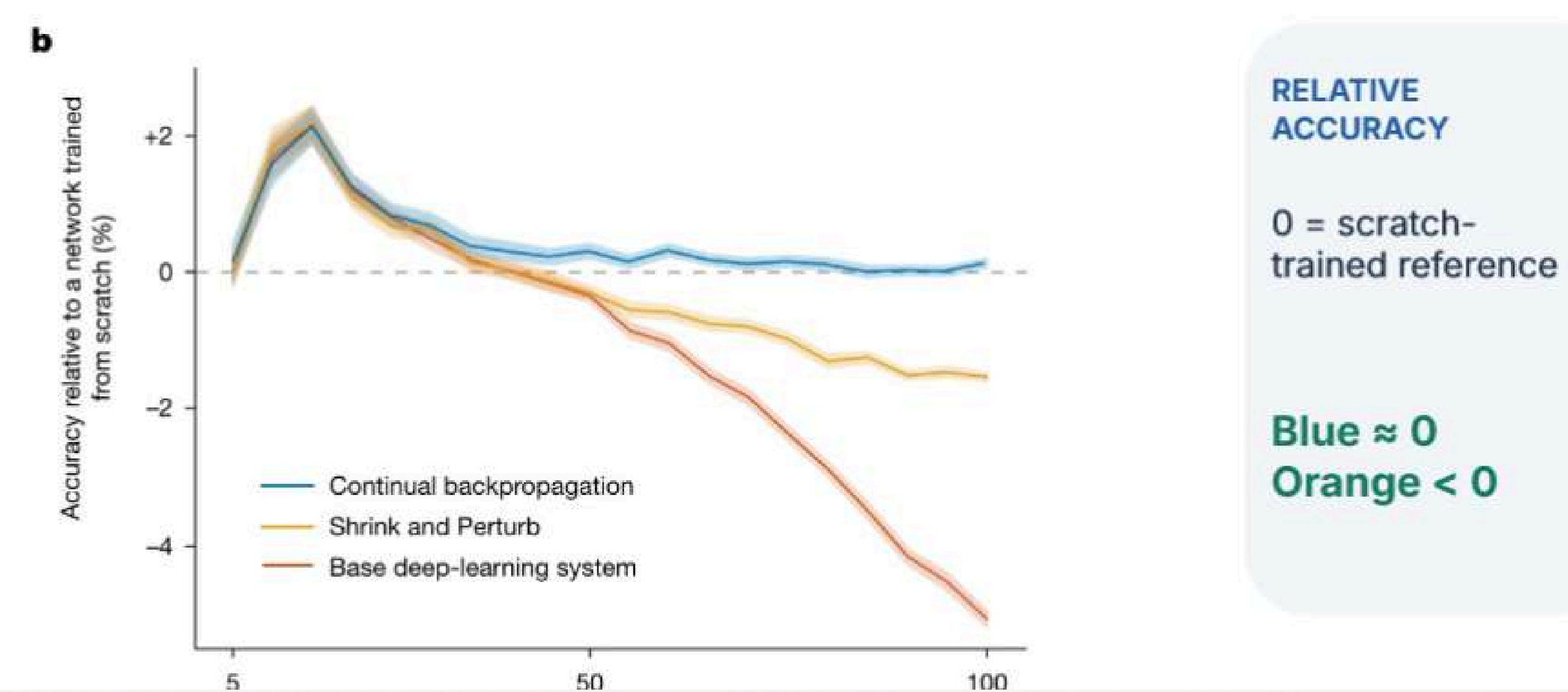
Y: test accuracy (%)

CBP performs best among these representative interventions

Source: Dohare et al., Nature 632 (2024), Fig. 1c

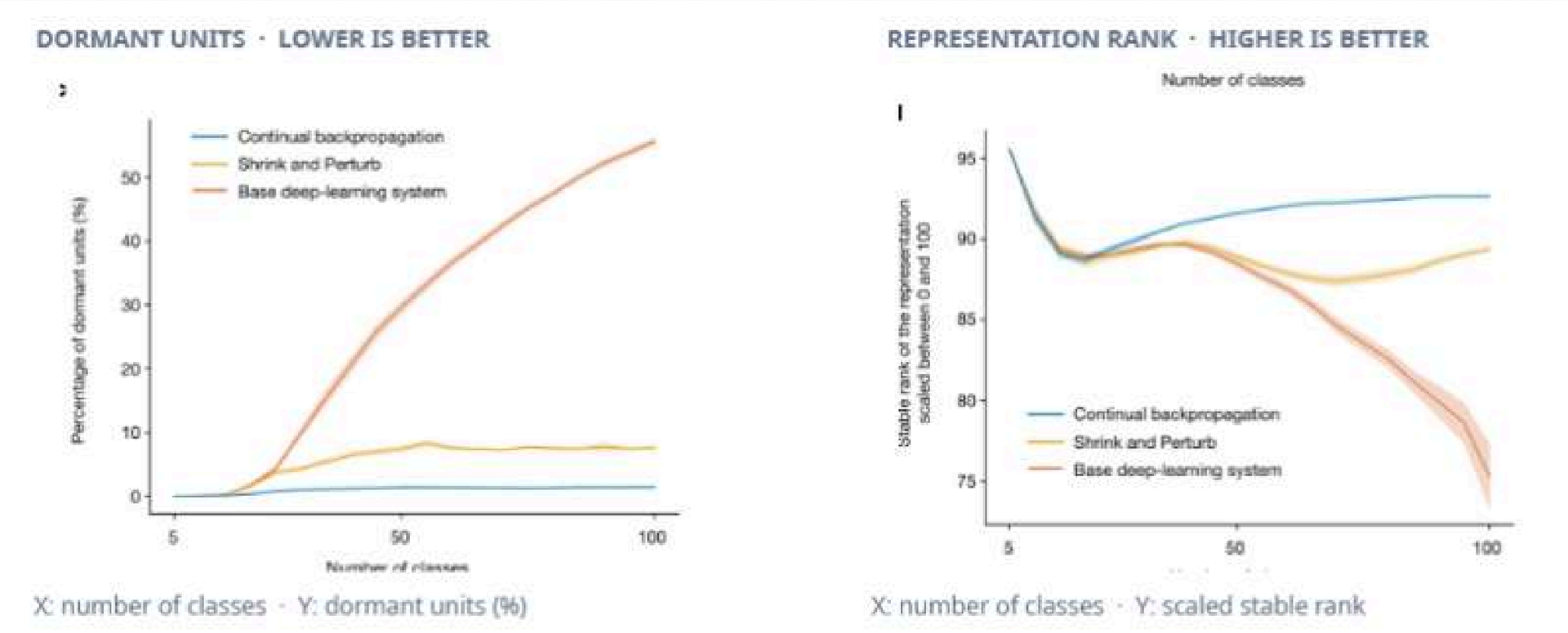
Experiment 2

CBP stays competitive with fresh retraining on CIFAR-100



Source: Dohare et al., Nature 632 (2024), Fig. 2b

CBP preserves active units and diverse CIFAR-100 features

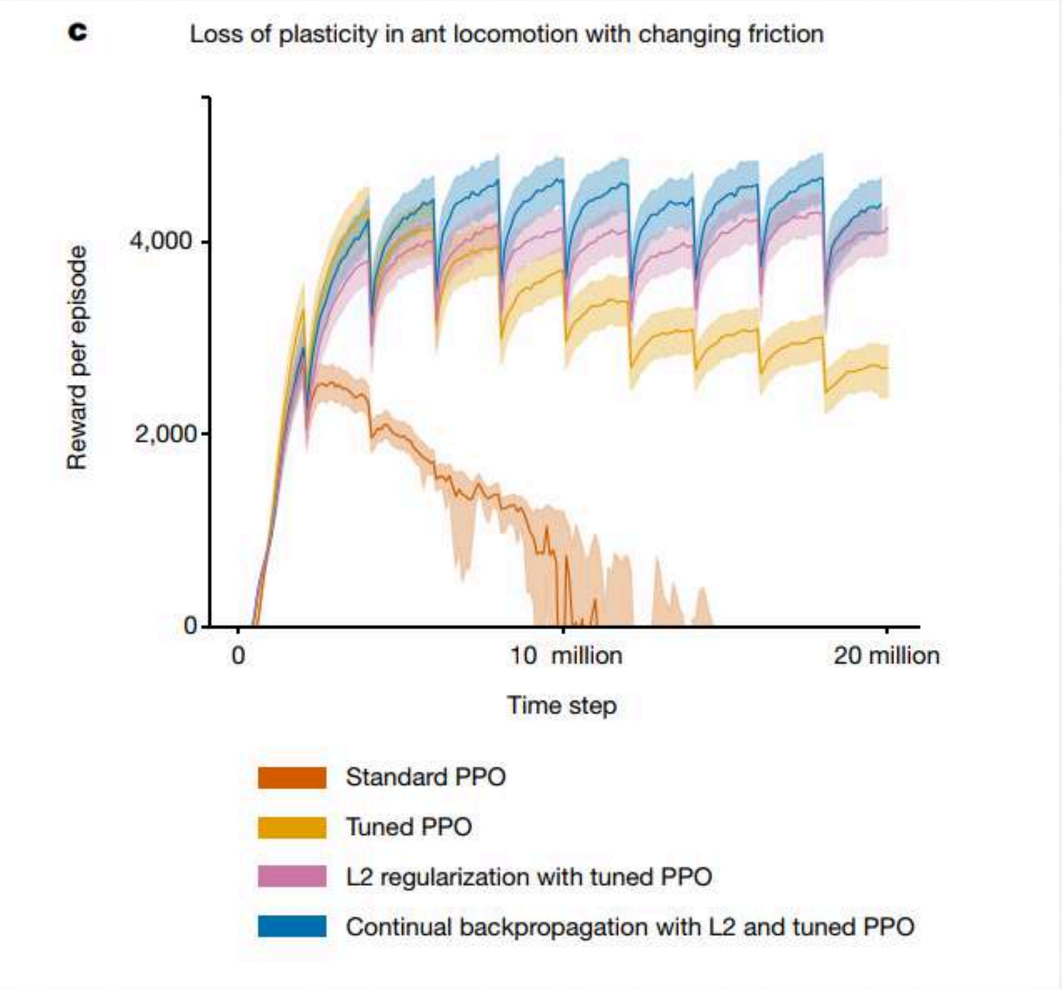


CBP maintains few dormant units and a higher-rank representation

Source: Dohare et al., Nature 632 (2024), Fig. 2c-d

Experiment 3

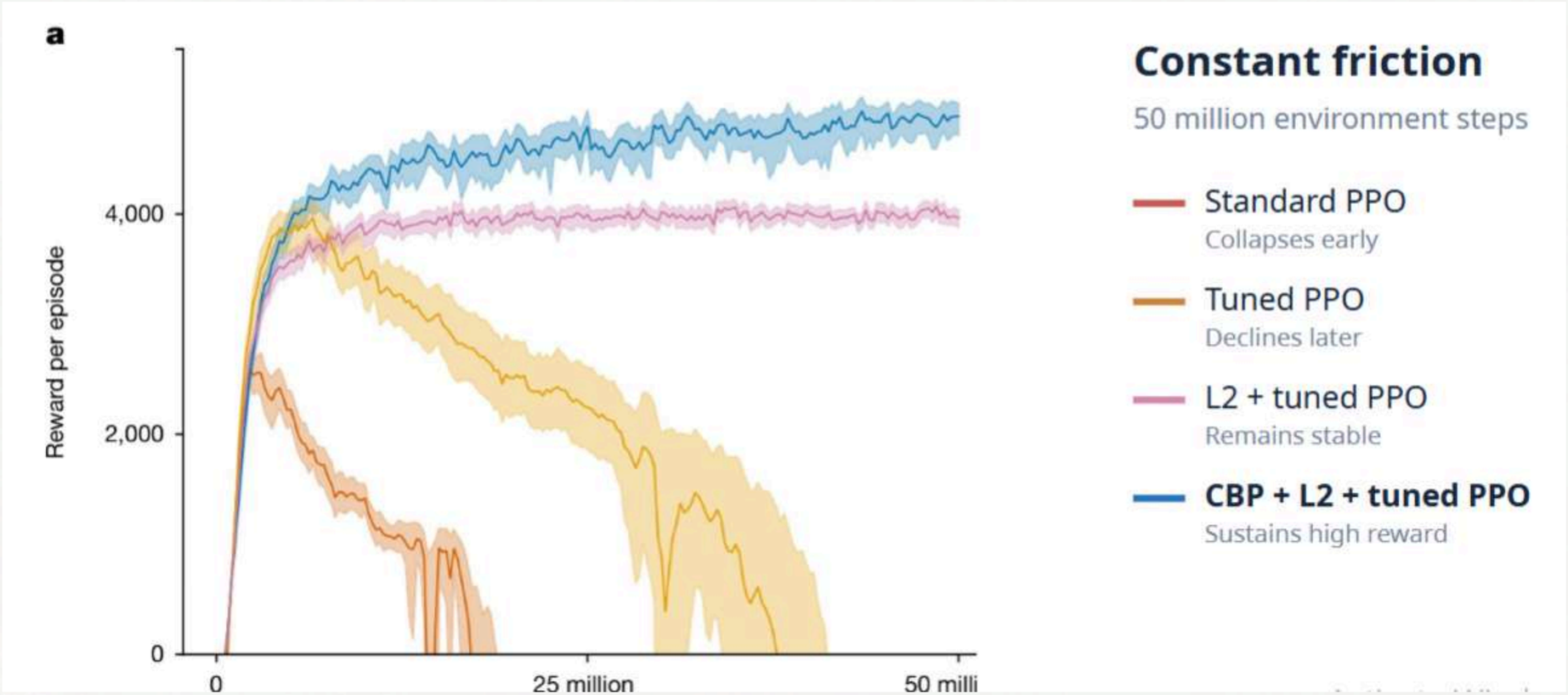
CBP + L2 preserves adaptation through repeated friction changes



Source: Dohare et al., Nature 632 (2024), Fig. 3c

Experiment 4

CBP + L2 sustains performance even without friction changes



Source: Dohare et al., Nature 632 (2024), Fig. 4a

Preserving plasticity does not guarantee memory retention

Remembering old tasks

Stability

Retention is not established

Learning new tasks

Plasticity

CBP imroves continued trainability

Low utility on today’s data does not mean a neuron was never useful

The paper's three objectives

01 — DEMONSTRATE
GRADUAL LOSS OF LEARNING ABILITY

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INTRODUCE CONTINUAL BACKPROPAGATION TO HELP PRESERVE THE NETWORK'S CAPACITY TO LEARN.

6.Key takesways & Discussion

1. Standard methods can lose the ability to learn new tasks.

2. Feature dormancy and diversity loss accompany deterioration

.

3. Selective renewal can maintain plasticity in tested settings

Plasticity research leaves fundamental questions for lifelong agents

Can one method prevent both forgetting and plasticity loss?

Is stochastic feature renewal necessary?

Can a network predict its impending loss of plasticity?

The End

THANK YOU FOR LISTENING <3

7. Assumptions, Limitations & Open Problems

Each experiment makes specific assumptions about continual learning

Continual ImageNet

Known boundaries

A reset output head separates successive binary tasks

CIFAR-100

Old data retained

All currently available classes remain in training

Ant locomotion

Policy changes data

Interaction and optimizer dynamics confound failure modes

The mechanism and large-scale generalization remain open questions

Causal mechanism

Are dead units, lower rank, and growing weights causes; or indicators?

Replacement choices

Which neurons, when, and how much renewal? Fixed rate and maturity parameters remain

Broader scaling

Do similar benefits carry over to very large models and less controlled task streams?

CBP leaves open questions about isolation, utility, and memory

CBP + L2: in ant locomotion CBP alone was parameter-sensitive

Local utility heuristic: does not measure global task importance

Retaining knowledge: no general forgetting guarantee

8.Subsequent research

Later work explores replay and model architecture as alternatives

Wang, Chandra & Zhang (2025)

Experience Replay Addresses Loss of Plasticity in Continual

Learning Replay combined with Transformers prevents plasticity loss in several tested tasks.

arXiv:2503.20018. A different explanation and experimental approach, not a universal guarantee

2026 studies refine both the mechanism and the replacement strategy

Prakash et al., ICML 2026
Spectral collapse and lost optimization directions

Hernandez-Garcia et al., CoLLAs 2026
Selective weight replacement can outperform unit replacement in some settings

Sources: PMLR 306 (2026); PMLR 330 (2026)